TDI EXCEL CAPSTONE ASSIGNMENT

DATA CLEANING & PREPARATION

The dataset underwent several cleaning and transformation steps to ensure accuracy, consistency, and suitability for analysis and dashboard creation.

A. Data Integration

The complaints dataset was enriched by merging it with the products and issues tables using **XLOOKUP** functions. This allowed product IDs and issue IDs to be replaced with meaningful descriptive fields such as product names, sub-products, issue, and sub-issue categories.

B. Standardization of State Names

State abbreviations were converted into full state names using a lookup table to improve readability and enhance the clarity of geographic analysis within the dashboard.

C. Creation of Calculated Fields

1. Response Time (Days)

A new column was created to calculate the number of days between complaint submission and receipt: date submitted subtracted from date received.

The column was formatted as a numerical value to represent the response time in days.

D. Handling Missing and Inconsistent Data

The dataset contained a significant proportion of "Unknown" values, particularly in the *Consumer Disputed* field. These were retained for transparency but highlighted as a limitation in the analysis.

E. Data Reduction for Visualization

Due to the high number of issue categories, a **Top 15 filter** was applied to focus on the most impactful complaint drivers. This ensured clarity and readability in visualizations while still capturing the majority of complaint volume.

F. Formatting and Structuring

- Converted date fields to proper date format
- Ensured categorical consistency (Yes/No fields)
- Structured the dataset into a clean table format for pivot analysis

Problem Statement Definition

Clearly define the problem that the Financial Consumer Complaints Analysis Dashboard aims to address.

The dashboard aims to answer the following questions;

- Where are the complaints coming from? (which state? Concerning which products?)
- What are customers complaining about ie what are the main issues?
- How well is the company responding?
- Why are some complaints unresolved or disputed?

Identify the key challenges facing Cumulus Financial based on what the data reveals, not what you assume going in.

1. Delays in complaint resolution as indicated by untimely responses
2. High levels of disputed complaints suggesting dissatisfaction with resolutions
3. Limited transparency due to low public response rates or missing consumer consent
4. Concentration of complaints around specific products or issue types
5. Variability in effectiveness across complaint submission channels
6. Geographic disparities in complaint volume and resolution outcomes
7. Inconsistent company response strategies impacting customer satisfaction

<u>Data Analysis</u>

What insights can you uncover from the data?

How do complaint trends vary by different parameters?

Complaint Trends:

1. Over Time
   - Complaints rise steadily from 2011 to ~2018, peaking around 3,200+
   - Slight stabilization after, but still consistently high

Insight:

- This isn't a one-off spike. It's a systemic growth in complaints
- Likely tied to:
  - Increased customer base
  - Or worsening service experience

2. By Product

Top offenders:

- Credit Card (~19,176)
- Checking/Savings (~13,436)

- Mortgage (~12,470)

Insight:

- Core financial products = highest friction points
- These are high-usage products meaning more exposure hence more failure points

3. By Issue

   Top issues:
   - Managing account (8,849)
   - Deposits & withdrawals (6,127)
   - Payment struggles (mortgage/loans ~3,500 range)

Insight: Complaints are not random
They cluster around money movement + account control

4. By Geography

Top states:

- California (12,107)
- New York (~6,800)
- Florida (~6,700)

Insight: High population states dominate as expected. These should be benchmarked against population for fairness.

5. By Channel

- Web dominates complaint intake
- Followed by Referral & Phone

Insight:

- Digital channels = primary complaint gateway
- But also:
  - Easier to complain online → higher volume

What is the relationship between how a complaint is resolved and whether the consumer disputes that resolution?

<u>Resolution vs Disputes</u>

Dispute Breakdown

- Unknown: 49% (huge data gap)
- No: 41%
- Yes: 10%

Insight: True dispute rate may be higher than 10%. Data quality issue is affecting conclusions

<u>Response Type vs Disputes</u>

Key pattern:

  - "Closed with explanation" → highest volume AND noticeable disputes
  - "Closed with monetary relief" → fewer disputes
  - "Closed without relief" → disputes present (as expected)

Insight:

- Explanations alone is not equal to satisfaction
- Customers want resolution, not storytelling

Are there patterns that might predict which complaint types are least likely to be resolved satisfactorily?

Timeliness vs Satisfaction

  - 98% timely response
  - BUT disputes still exist

Insight:

- Speed does not always mean customer satisfaction
- Company is fast but not always effective

<u>Complaint Patterns and Resolution Insights</u>
Identify high-complaint segments and analyze the potential factors contributing to poor resolution outcomes or consumer disputes.

High-Complaint + High-Risk Segments

Segment 1:

Credit Card + Managing Account / Billing Issues

- Highest volume

- High dispute exposure

Segment 2:

Bank Accounts + Deposits/Withdrawals

- Operational failures (money movement errors)
- Trust-sensitive → high dissatisfaction risk

Segment 3:

Mortgage + Payment Difficulty

- Financial stress situations
- Customers expect solutions, not explanations

Provide recommendations based on the trends you observe to improve how complaints are handled and prevented.

Reduce "Explanation-Only" Resolutions

Data says: High volume + associated disputes

Action:

- Introduce policy: Require actionable resolution for high-risk issues
- Track: % of complaints resolved with relief

Focus on Money-Movement Issues First

Data says: Deposits, withdrawals, account management dominate complaints

Action:

- Audit:
    - Transaction systems
    - Reversal processes
- Create: Dedicated rapid-response team for these issues

Improve Data Quality

Data says: 49% dispute status = unknown

Action:

- Make dispute field mandatory
- Clean historical data where possible

Without this, decision-making is partially blind

Segment-Based Support Teams

Data says: Credit cards, bank accounts, mortgages dominate

Action:

- Create specialized resolution teams per product
- Train agents in product-specific issue handling

Prioritize High-Volume States

Data says: CA, NY, FL lead complaints

Action:

- Allocate:
    - More support staff
    - Localized service improvements

Channel Optimization

Data says: Web = highest complaints

Action:

- Improve:
    - Self-service resolution tools
    - FAQ systems
- Add:
    - Smart triaging before submission

Shift KPI Focus

Current: Timely response = 98%

New KPIs:

- Resolution satisfaction rate
- Dispute rate by response type
- Average resolution effectiveness

Because: Fast bad service is still bad service

DASHBOARD FEATURES

An interactive dashboard was developed in Microsoft Excel to provide a comprehensive view of consumer complaint patterns, resolution performance, and customer satisfaction indicators.

A. Key Performance Indicators (KPIs)

The dashboard includes four primary KPI cards:

- Total Complaints (75,513)
- % Timely Responses (98.05%)
- % Disputed Complaints (9.75%)
- Average Response Time (3 days)

Purpose: These KPIs provide a quick snapshot of overall performance and service efficiency.

B. Trend Analysis

A time-series chart visualizes complaint volumes over time (quarterly), enabling identification of growth patterns, peaks, and fluctuations in complaint activity.

C. Product-Level Analysis

A bar chart displays the number of complaints by product category, highlighting high-risk product segments such as credit cards and banking services.

D. Issue Analysis

A ranked bar chart of the **Top 15 issues** identifies the most frequent customer concerns, enabling targeted root-cause analysis.

E. Channel Performance Analysis

A clustered bar chart compares complaint volumes and dispute outcomes across different submission channels (Web, Phone, Referral, etc.), highlighting variations in performance.

F. Geographic Analysis

A bar chart of the top 10 states with the highest complaint volumes provides insights into regional patterns and potential areas requiring operational focus.

G. Resolution Performance

- Pie chart for **Timely vs Untimely responses**
- Donut chart for **Disputed vs Non-disputed complaints**

Purpose: To evaluate both operational efficiency and customer satisfaction.

H. Response Effectiveness Analysis

A comparative chart shows how different resolution types (e.g., closed with explanation, monetary relief) relate to dispute outcomes, providing insight into resolution quality.

I. Interactivity (Slicers)

The dashboard includes slicers for:

- State
- Submission Channel
- Timely Response

Purpose: These allow users to dynamically filter the data and explore patterns across different segments.

J. User Experience Design

The dashboard was designed with a clean and structured layout, grouping related insights together and using consistent color schemes to enhance readability and usability.

This dashboard enables stakeholders to move from high-level performance metrics to detailed root-cause analysis, supporting informed decision-making and targeted service improvement strategies.